Chapter 15

Restriction and Induction of Representations

Throughout, G is a finite group and $H \leqslant G$ a subgroup.

15.1 Restriction

Definition 15.1.1: Restriction

Given a representation V of G, the restriction of V to H, denoted $\mathrm{Res}_H^G V$, is V viewed as a representation of H (forget the action of elements outside H).

This defines a functor $\mathrm{Res}_H^G : \mathrm{Rep}(G) \to \mathrm{Rep}(H)$, where the categories have objects = finite-dimensional complex representations and morphisms = G-equivariant (resp. H-equivariant) linear maps.

An irreducible representation of G need not remain irreducible upon restriction to H. The problem we address next is: can we go in the opposite direction?

15.2 Induced Representations

Suppose V is a representation of G and $W \subset V$ is a subspace that is invariant under H (so W is a subrepresentation of $\mathrm{Res}_H^G V$), but not necessarily under all of G. For $g \in G$, the subspace $gW \subset V$ depends only on the coset gH, and gW is a representation of the conjugate subgroup gHg^{-1} via $H \xrightarrow{\rho} \mathrm{GL}(W)$, $gHg^{-1} \xrightarrow{\sim} \mathrm{GL}(gW)$ (conjugation by g).

The simplest scenario is when

$$V = \bigoplus_{\sigma \in G/H} \sigma W$$

as a direct sum of vector spaces, where σ ranges over a set of coset representatives. In this case, $\dim V = [G : H] \cdot \dim W$, and the entire representation V is completely determined by the H-representation W.

Definition 15.2.1: Induced representation

A representation V of G, together with a subspace $W \subset V$ invariant under H (i.e. W is a subrepresentation of $\mathrm{Res}_H^G V$), is said to be induced by $W \in \mathrm{Rep}(H)$ if

$$V = \bigoplus_{\sigma \in G/H} \sigma W.$$

We write $V = \mathrm{Ind}_H^G W$. Concretely: fix coset representatives $\sigma_1, \ldots, \sigma_k$ of G/H (where $k = [G : H]$). Then every $v \in V$ can be written uniquely as $v = \sigma_1 w_1 + \cdots + \sigma_k w_k$ with $w_i \in W$, and G acts by permuting the summands: if $g\sigma_i = \sigma_j h$ for some $h \in H$, then $g(\sigma_i w) = \sigma_j(hw)$.

Theorem 15.2.1 Existence and uniqueness of induced representations

Given a representation W of H, the induced representation $V = \mathrm{Ind}_H^G W$ exists and is unique up to isomorphism of G-representations.

Proof: Uniqueness. Suppose V is a representation of G and $W \subset V$ is H-invariant with $V = \bigoplus_{i=1}^k \sigma_i W$. For any $g \in G$, there is a unique j such that $g\sigma_i \in \sigma_j H$, i.e. $h = \sigma_j^{-1} g \sigma_i \in H$, and necessarily $g(\sigma_i w) = \sigma_j(hw)$. This determines the G-action on V uniquely from the H-action on W.

Existence. Build $V = \bigoplus_{i=1}^k \sigma_i W$ where the σ_i are formal symbols (i.e. V consists of k copies of W as a vector space). Define $g \in G$ to act by: given $g\sigma_i = \sigma_j h$, set $g(\sigma_i w) = \sigma_j(hw)$. This is well-defined (independent of the choice of coset representatives) and defines a group homomorphism $G \to \mathrm{GL}(V)$. $\square$

15.2.1 Examples

Example 15.2.1 (Permutation representation from induction)

The permutation representation of G on the coset space G/H equals $\mathrm{Ind}_H^G U$, where U is the trivial representation of H. Indeed, the permutation representation has basis $\{e_{\sigma H}\}_{\sigma \in G/H}$, the element e_H spans a 1-dimensional subspace fixed by H, and $gW = \mathrm{span}(e_{gH})$, so $V = \bigoplus_{gH \in G/H} gW$.

Example 15.2.2 (Regular representation from induction)

The regular representation of G is $\mathrm{Ind}_H^G R_H$, where R_H is the regular representation of H. Here $W = \mathrm{span}\{e_h : h \in H\} \subset V = \mathrm{span}\{e_g : g \in G\}$.

15.2.2 Properties of Induction

Proposition 15.2.1 Basic properties

Let $H \leqslant G$, and let W, W' be representations of H and U a representation of G.

(i) $\mathrm{Ind}_H^G(W \oplus W') \cong \mathrm{Ind}_H^G(W) \oplus \mathrm{Ind}_H^G(W')$.

(ii) $\mathrm{Ind}_H^G(\mathrm{Res}_H^G(U) \otimes W) \cong U \otimes \mathrm{Ind}_H^G(W)$.

(iii) In particular, $\mathrm{Ind}_H^G(\mathrm{Res}_H^G(U)) \cong U \otimes \mathrm{Ind}_H^G(U_H)$, where U_H is the trivial representation of H. The right side is U tensored with the permutation representation on G/H.

Proof of (ii): Write $\mathrm{Ind}_H^G(W) = \bigoplus_{\sigma \in G/H} \sigma W$. Then

$$U \otimes \mathrm{Ind}_H^G(W) = \bigoplus_{\sigma \in G/H} (U \otimes \sigma W) = \bigoplus_{\sigma \in G/H} \sigma(U \otimes W),$$

where $U \otimes W \subset U \otimes \mathrm{Ind}_H^G(W)$ is invariant under H, and as an H-representation it is $\mathrm{Res}_H^G(U) \otimes W$. $\square$

Note:-

In general, $\mathrm{Ind}(W \otimes W') \not\cong \mathrm{Ind}(W) \otimes \mathrm{Ind}(W')$.

15.2.3 Character of an Induced Representation

Proposition 15.2.2 Character formula for induction

Let W be a representation of $H \leqslant G$. Then

$$\chi_{\mathrm{Ind}_H^G W}(g) = \frac{1}{|H|} \sum_{\substack{s \in G \\ s^{-1} g s \in H}} \chi_W(s^{-1} g s).$$

Proof: Choose coset representatives $\sigma_1, \ldots, \sigma_k$ of G/H. The element $g \in G$ maps $\sigma_i W$ to $\sigma_j W$ where j is determined by $g\sigma_i \in \sigma_j H$. If $i \neq j$, this contributes nothing to the trace. If $i = j$, then $h = \sigma_i^{-1} g \sigma_i \in H$ and g acts on $\sigma_i W$ as $\sigma_i h \sigma_i^{-1}$ composed with the identification $\sigma_i W \cong W$, so the trace on $\sigma_i W$ is $\mathrm{Tr}(h|_W) = \chi_W(\sigma_i^{-1} g \sigma_i)$. Summing over all σ_i:

$$\chi_{\mathrm{Ind}\, W}(g) = \sum_{\substack{i \\ \sigma_i^{-1} g \sigma_i \in H}} \chi_W(\sigma_i^{-1} g \sigma_i).$$

Since each coset $\sigma_i H$ has $|H|$ elements, and the condition $\sigma_i^{-1} g \sigma_i \in H$ is the same for any representative of the coset, we can rewrite this as a sum over all $s \in G$ with the normalization factor $1/|H|$. $\square$

15.3 Frobenius Reciprocity

Theorem 15.3.1 Frobenius reciprocity

Let U be a representation of G and W a representation of $H \leqslant G$. Then every H-equivariant map $\varphi : W \to \mathrm{Res}_H^G(U)$ extends uniquely to a G-equivariant map $\tilde{\varphi} : \mathrm{Ind}_H^G(W) \to U$. That is,

$$\mathrm{Hom}_H(W, \mathrm{Res}_H^G U) \cong \mathrm{Hom}_G(\mathrm{Ind}_H^G W, U).$$

Proof: Choose coset representatives $\sigma_1, \ldots, \sigma_k \in G$ and write $V = \mathrm{Ind}_H^G W = \bigoplus_{i=1}^k \sigma_i W$.

Uniqueness. Given $\varphi : W \to \mathrm{Res}(U)$, if $\tilde{\varphi} : V \to U$ is G-equivariant with $\tilde{\varphi}|_W = \varphi$, then for any $\sigma_i w \in \sigma_i W$:

$$\tilde{\varphi}(\sigma_i w) = \sigma_i \cdot \tilde{\varphi}(w) = \sigma_i \cdot \varphi(w),$$

where the last σ_i acts via the G-action on U. This determines $\tilde{\varphi}$ on each summand, hence on all of V.

Existence and G-equivariance. Define $\tilde{\varphi}(\sigma_i w) = \sigma_i \cdot \varphi(w)$. To verify G-equivariance: let $g \in G$ act on V by sending $\sigma_i W$ to $\sigma_j W$ where $g\sigma_i = \sigma_j h$ for $h \in H$. For $\sigma_i w \in \sigma_i W$:

$$\tilde{\varphi}(g(\sigma_i w)) = \tilde{\varphi}(\sigma_j(hw)) = \sigma_j \cdot \varphi(hw) = \sigma_j \cdot h \cdot \varphi(w) = g \cdot \sigma_i \cdot \varphi(w) = g \cdot \tilde{\varphi}(\sigma_i w),$$

using H-equivariance of φ in the third equality and $\sigma_j h = g\sigma_i$ in the fourth. So $\tilde{\varphi}$ is G-equivariant.

Conversely, given $\tilde{\varphi} \in \mathrm{Hom}_G(V, U)$, its restriction to $W \subset V$ is H-equivariant. These two maps are inverse to each other.

Comparing dimensions of the Hom spaces gives: $\square$

Corollary 15.3.1 Frobenius reciprocity (character form)

For any representation U of G and W of H,

$$\langle \chi_{\mathrm{Ind}_H^G W}, \chi_U \rangle_G = \langle \chi_W, \chi_{\mathrm{Res}_H^G U} \rangle_H.$$

Note:-

Thus, if U is an irreducible representation of G and W an irreducible representation of H, the number of times W appears in $\mathrm{Res}_H^G(U)$ equals the number of times U appears in $\mathrm{Ind}_H^G(W)$.

15.4 Example: $S_4 \supset S_3$

Example 15.4.1 (Restriction and induction between S_4 and S_3)

Let $G = S_4$ with irreducibles U_4, U_4', V_4, V_4', W (dimensions 1, 1, 3, 3, 2), and $H = S_3$ with irreducibles U_3, U_3', V_3 (dimensions 1, 1, 2). We compute restrictions using character tables (restricting the character of a representation of S_4 to elements of $S_3 \hookrightarrow S_4$):

- $\mathrm{Res}(U_4) = U_3$ and $\mathrm{Res}(U_4') = U_3'$ (trivial and sign restrict to trivial and sign).
- $\mathrm{Res}(V_4) = V_3 \oplus U_3$. Indeed, the permutation representation $\mathbb{C}^4$ restricts as $\mathbb{C}^3 \oplus \mathbb{C}$ (the 4th coordinate is fixed by S_3), so $\mathrm{Res}(V_4 \oplus U_4) = V_3 \oplus U_3 \oplus U_3$.
- $\mathrm{Res}(V_4') = V_3 \oplus U_3'$ (since $V_4' = V_4 \otimes U_4'$ and $V_3 \otimes U_3' \cong V_3$).
- $\mathrm{Res}(W) = V_3$. The representation W factors through $S_4/\{e, (12)(34), (13)(24), (14)(23)\} \cong S_3$, and W as a representation of this quotient is the standard representation V_3.

By Frobenius reciprocity, we can read off the induced representations:

- $\mathrm{Ind}_{S_3}^{S_4}(V_3)$ contains V_4, V_4', W each with multiplicity 1, giving $\mathrm{Ind}(V_3) = V_4 \oplus V_4' \oplus W$ (dim $= 3 + 3 + 2 = 8 = [S_4 : S_3] \cdot 2$).
- $\mathrm{Ind}_{S_3}^{S_4}(U_3) = U_4 \oplus V_4$ (dim $= 1 + 3 = 4 = [S_4 : S_3] \cdot 1$).
- $\mathrm{Ind}_{S_3}^{S_4}(U_3') = U_4' \oplus V_4'$ (dim $= 1 + 3 = 4$).

Exercise 15.4.1 Exercises

(1) Let $H = \{e\} \leqslant G$. Show that $\mathrm{Ind}_{\{e\}}^G W \cong W \otimes R$ where R is the regular representation of G and W is viewed as a vector space with trivial G-action on the second factor.

(2) Let $H = A_n \leqslant S_n$ for $n \geqslant 2$. Show that $\mathrm{Ind}_{A_n}^{S_n}(U)$, where U is the trivial representation of A_n, is isomorphic to $U_{S_n} \oplus U_{S_n}'$ (the trivial plus the sign representation of S_n).

(3) Using Frobenius reciprocity and the character table of S_5, compute $\mathrm{Ind}_{S_4}^{S_5}(V_4)$ (where V_4 is the standard representation of S_4, embedded as the subgroup fixing 5).

(4) Prove that the character formula for induction can alternatively be written as $\chi_{\mathrm{Ind}\, W}(g) = \sum_{i : \sigma_i^{-1} g \sigma_i \in H} \chi_W(\sigma_i^{-1} g \sigma_i)$ where $\sigma_1, \ldots, \sigma_k$ are coset representatives.

(5) (Transitivity of induction) If $K \leqslant H \leqslant G$, show $\mathrm{Ind}_H^G \mathrm{Ind}_K^H W \cong \mathrm{Ind}_K^G W$.

Solution

(1) $\mathrm{Ind}_{\{e\}}^{G} W = \bigoplus_{g \in G} gW$, a direct sum of $|G|$ copies of W, with G permuting the summands. This is exactly $W \otimes R$ where the G-action is on the R factor.

(2) $\mathrm{Ind}_{A_n}^{S_n}(U) = \bigoplus_{\sigma \in S_n/A_n} \sigma U$. Since $[S_n : A_n] = 2$, this is 2-dimensional with basis e_1, e_2 indexed by the two cosets. Even permutations fix both; odd permutations swap them. The invariant subspace $\mathrm{span}(e_1 + e_2) \cong U_{S_n}$, and the anti-invariant $\mathrm{span}(e_1 - e_2) \cong U'_{S_n}$.

(3) First compute $\mathrm{Res}_{S_4}^{S_5}$ of each irreducible of S_5 and find the multiplicity of V_4. From Frobenius reciprocity, the multiplicity of an S_5-irreducible X in $\mathrm{Ind}(V_4)$ equals the multiplicity of V_4 in $\mathrm{Res}(X)$. Computing (by restricting characters to S_4): $\mathrm{Ind}_{S_4}^{S_5}(V_4) = V \oplus V' \oplus W \oplus W' \oplus \bigwedge^2 V$ ($\dim = 4 + 4 + 5 + 5 + 6 = 24$... but $[S_5 : S_4] \cdot 3 = 15$). Let me recompute: $[S_5 : S_4] = 5$ and $\dim V_4 = 3$, so $\dim \mathrm{Ind}(V_4) = 15$. The correct decomposition requires careful character computation.

(4) The sum $\sum_{i:\sigma_i^{-1} g \sigma_i \in H} \chi_W(\sigma_i^{-1} g \sigma_i)$ counts each valid conjugate exactly once per coset. In the formula $\frac{1}{|H|} \sum_{s \in G, s^{-1} g s \in H}$, replacing s by $\sigma_i h$ for $h \in H$: the condition $s^{-1} g s \in H$ becomes $h^{-1}(\sigma_i^{-1} g \sigma_i) h \in H$, which holds iff $\sigma_i^{-1} g \sigma_i \in H$. The $|H|$ values of h give $\chi_W(h^{-1}(\sigma_i^{-1} g \sigma_i) h) = \chi_W(\sigma_i^{-1} g \sigma_i)$ (class function), contributing $|H|$ times the same value. Dividing by $|H|$ recovers the coset-representative sum.

(5) Write coset representatives for G/K by combining those for G/H and H/K: if $\{\sigma_i\}$ represents G/H and $\{\tau_j\}$ represents H/K, then $\{\sigma_i \tau_j\}$ represents G/K. The direct sum decompositions are compatible: $\mathrm{Ind}_K^G W = \bigoplus_{i,j} \sigma_i \tau_j W = \bigoplus_i \sigma_i (\bigoplus_j \tau_j W) = \bigoplus_i \sigma_i (\mathrm{Ind}_K^H W) = \mathrm{Ind}_H^G(\mathrm{Ind}_K^H W)$.